

SHUFANG ZHU

Lecturer (Assistant Professor)
School of Computer Science and Informatics
University of Liverpool
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EDUCATION

PhD in SOFTWARE ENGINEERING East China Normal University , China Advisor: Prof. Geguang Pu, Co-advisor: Prof. Moshe Y. Vardi Thesis: Program Synthesis of Linear Temporal Logic over Finite Traces	Sept. 14 - Mar. 20
BSc in SOFTWARE ENGINEERING East China Normal University , China	Sept. 10 - Jun. 14

PREVIOUS ACADEMIC APPOINTMENT

Senior Research Associate in DEPARTMENT OF COMPUTER SCIENCE University of Oxford , UK Mentor: Prof. Giuseppe De Giacomo	Feb. 23 - Aug. 24
Research Associate in DEPT. OF COMP., CONTROL & MANAGEMENT ENGINEERING Sapienza University of Rome , Italy Mentor: Prof. Giuseppe De Giacomo	Dec. 20 - Nov. 22

RESEARCH INTERESTS

My expertise lies in the interdisciplinary research area of artificial intelligence (AI) and formal methods (FM), with a focus on **automated planning and synthesis**. My research vision is to advance the development of **trustworthy** autonomous AI systems through trustworthy-by-design techniques.

GRANTS

Royal Society International Exchanges Grant "Exploiting Knowledge Representation in Boolean Functional Synthesis for Temporal Synthesis", £12,000 PI: Dr. Shufang Zhu, Co-PI: Prof. Supratik Chakraborty (IIT Bombay)	Dec. 25 - Dec. 27
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AWARDS AND HONORS

Panelist at KRPlan (joint workshop at KR and ICAPS)	Nov. 25
 Best Paper Award at CIAA 2025	Sept. 25
Invited to Dagstuhl Seminar on Knowledge Compilation in Artificial Intelligence, Databases, and Formal Methods	May. 26
Invited to Dagstuhl Seminar on Software Contracts Meet System Contracts	Jan. 26
Invited Speaker at the 7th edition of the International Symposium on AI Verification (SAIV)	Jul. 24
Invited to Dagstuhl Seminar on Automated Synthesis: Functional, Reactive and Beyond	Apr. 24
Selected Mentee at F+Cube Program TU Delft, The Netherlands	2023

Future Digileader Digital Futures, Sweden	2023
Rising Star in Electrical Engineering and Computer Science (EECS) UT Austin, USA	2022
Invited to Dagstuhl Seminar on The Futures of Reactive Synthesis	Sept. 23
Chinese Government Scholarship Chinese Scholarship Council	May. 2016
Academic Scholarship East China Normal University	2015, 2016, 2017, 2018, 2019

INVITED TALKS

- **Trustworthy-by-Design Autonomous AI Systems**, In the 7th International Symposium on AI Verification (SAIV), 23/07/2024, Montreal, Canada.

Invited Tutorials

- **LTL_f Synthesis and Planning Under Unreliable Input**, International Conference on Automated Planning and Scheduling (ICAPS) , 2025

RESEARCH COMMUNITY SERVICES

Organizing Committee

Workshop Chair. CONFEST 2026

Co-Chair. the 14th Synthesis (SYNT) Workshop, co-located with CAV 2025

Co-Chair. ECAI 2024 Workshop: Highlights of Reasoning about Actions, Planning and Reactive Synthesis

Co-Chair. AAAI Spring Symposium Series 2023: On the Effectiveness of Temporal Logics on Finite Traces

Program Committee

2027. AAAI, FSE

2026. AAAI, KR, TASE, IJCAI

2025. AAAI, SPIN, ECAI, KR, SAIV

2024. IJCAI, AAAI, KR, AAMAS, FM, FMCAD, CAV Artifact Evaluation, SAIV

2023. IJCAI, KR, FMCAD, ECAI

2022. AAAI, IJCAI

2021. AAAI

Journal Reviewer

2026. Philosophical Transactions A

2025. Transactions on Software Engineering, International Journal on Software Tools for Technology Transfer

2024. Logical Methods in Computer Science, Autonomous Agents and Multi-Agent Systems, Science China Information Sciences

2023. Artificial Intelligence Journal

2020. Mathematical Problems in Engineering, IEEE Access

2017. Formal Methods in System Design

TEACHING

Lecturer, Multi-Agent Systems (UG/MS level)
University of Liverpool, UK

Second Semester. 27 & 26

Guest Lecturer, One lecture on 'Symbolic Synthesis Techniques'
In the course "Foundations of Self-Programming Agents" (MS/PhD level)
University of Oxford, UK

Hilary Term. 24 & 23

Class Tutor, Foundations of Self-Programming Agents (MS/PhD level)
University of Oxford, UK

Hilary Term. 24 & 23

Co-lecturer, Game-Theoretic Approach to Planning and Synthesis (MS/PhD level)
European Summer School on Artificial Intelligence ESSAI, Ljubljana, Slovenia

Jul. 24 & Jul. 23

Co-lecturer, Game-Theoretic Approach to Planning and Synthesis (MS/PhD level)
Italian PhD program in Artificial Intelligence & Artificial Intelligence Doctoral Academy
Sapienza University of Rome, Italy

Jul. 22

SUPERVISION

PhD students

1. [Harsha Chinta](#) (University of Liverpool) Dec. 25 - Present
2. [Yujian Cao](#) (University of Liverpool) Sept. 25 - Present

MSc students

1. Gianmarco Parretti (Sapienza University of Rome) Sept. 21 - Jul. 22
Thesis: Symbolic best-effort synthesis for specifications in Linear Temporal Logic on finite traces
Thesis received 110 (with Honors)/110 points

PUBLICATIONS

*** indicates author list has been sorted alphabetically by last name**

39. **[IJCAI-26]*** [Optimal \$LTL_f\$ Synthesis](#)
Yujian Cao, Sven Schewe, Qiyi Tang, **Shufang Zhu**
In Proc. of International Joint Conference on Artificial Intelligence (IJCAI) 2026
38. **[TASE-26]** [Incremental Reinforcement Learning with Temporally Dependent Goals](#)
Yi Yang, **Shufang Zhu**, Giuseppe De Giacomo, Qin Zhao, Xinchao Li and Dongdong An
In Proc. of International Symposium on Theoretical Aspects of Software Engineering (TASE) 2026
37. **[CAV-26]*** 🏆 [Fast Obligation Translation and Synthesis](#)
Alexandre Duret-Lutz, Giuseppe De Giacomo, Marcin Jurdzinski, Nir Piterman, Moshe Y. Vardi, **Shufang Zhu**
In Proc. of International Conference on Computer Aided Verification (CAV) 2026
36. **[KR-26]*** [On-the-fly \$LTL_f\$ Synthesis under Partial Observability](#)
Nadav Alon, Supratik Chakraborty, Alexandre Duret-Lutz, Dror Fried, Lucas Martinelli Tabajara, Moshe Y. Vardi, **Shufang Zhu**
In Proc. of International Conference on Principles of Knowledge Representation and Reasoning (KR) 2026
35. **[FoSSaCS-26]*** [The Complexity of Games with Randomised Control](#)
Sarvin Bahmani, Rasmus Ibsen-Jensen, Soumyajit Paul, Sven Schewe, Friedrich Slivovsky, Qiyi Tang, Dominik Wojtczak, **Shufang Zhu**
In Proc. of International Conference on Foundations of Software Science and Computation Structures (FoSSaCS) 2026
34. **[TOSEM-25]** [An On-the-Fly Synthesis Framework for \$LTL\$ over Finite Traces](#)
Shengping Xiao, Yongkang Li, **Shufang Zhu**, Jun Sun, Jianwen Li, Geguang Pu, Moshe Y. Vardi
ACM Transactions on Software Engineering and Methodology (TOSEM) 2025

33. **[OVERLAY-25]** [Learning to Coordinate without Communication under Incomplete Information](#)
Shenghui Chen, **Shufang Zhu**, Giuseppe De Giacomo, Ufuk Topcu
In Proc. of International Workshop on Artificial Intelligence and fOrmal VERification, Logic, Automata, and sYnthesis (OVERLAY) 2025
32. **[ICAPS-25]*** [LTL_f Adaptive Synthesis for Multi-Tier Goals in Nondeterministic Domains](#)
Giuseppe De Giacomo, Gianmarco Parretti, **Shufang Zhu**
In Proc. of International Conference on Automated Planning and Scheduling (ICAPS) 2025
31. **[ECAI-25]** [A Compositional Framework for On-the-Fly LTL_f Synthesis](#)
Yongkang Li, Shengping Xiao, **Shufang Zhu**, Jianwen Li, Geguang Pu
In Proc. of the European Conference on Artificial Intelligence (ECAI) 2025
30. **[KR-25]*** [Emerson-Lei and Manna-Pnueli Games for LTL_f⁺ and PPLTL⁺ Synthesis](#)
Daniel Hausmann, **Shufang Zhu**, Gianmarco Parretti, Christoph Weinhuber, Giuseppe De Giacomo, Nir Piterman
In Proc. of International Conference on Principles of Knowledge Representation and Reasoning (KR) 2025
29. **[CIAA-25]**  [Engineering an LTL_f Synthesis Tool](#)
Alexandre Duret-Lutz, **Shufang Zhu**, Nir Piterman, Giuseppe De Giacomo, Moshe Y. Vardi
In Proc. of International Conference on Implementation and Application of Automata (CIAA) 2025
28. **[SN Comput. Sci.-25]*** [Symbolic LTL_f Synthesis: A Unified Approach for Synthesizing Winning, Dominant, and Best-Effort Strategies](#)
Giuseppe De Giacomo, Gianmarco Parretti, **Shufang Zhu**
SN Computer Science, Vol. 6(2), 2025
27. **[Inf. Comput.-25]*** [LTL_f synthesis under environment specifications for reachability and safety properties](#)
Benjamin Aminof, Giuseppe De Giacomo, Antonio Di Stasio, Hugo Francon, Sasha Rubin, **Shufang Zhu**
Information and Computation, Vol. 303, Article 105255, 2025
26. **[TACAS-25]** [LydiaSyft: A Compositional Symbolic Synthesis Framework for LTL_f Specifications](#)
Shufang Zhu and Marco Favorito
In Proc. of International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS) 2025
25. **[AAAI-24]*** [Mimicking Behaviors in Separated Domains \(Abstract Reprint\)](#)
Giuseppe De Giacomo, Dror Fried, Fabio Patrizi, Shufang Zhu
In Proc. of AAAI Conference on Artificial Intelligence (AAAI) 2024
24. **[FM-24]** [Misconceptions in Finite-Trace and Infinite-Trace Linear Temporal Logic](#)
Ben Greenman, Siddhartha Prasad, Antonio Di Stasio, **Shufang Zhu**, Giuseppe De Giacomo, Shriram Krishnamurthi, Marco Montali, Tim Nelson, Milda Zizyte
In Proc. of the International Symposium on Formal Methods (FM) 2024
23. **[IJCAI-24]** [The Trembling-Hand Problem for LTL_f Planning](#)
Pian Yu, **Shufang Zhu**, Giuseppe De Giacomo, Marta Kwiatkowska, Moshe Y. Vardi
In Proc. of the International Joint Conference on Artificial Intelligence (IJCAI) 2024
22. **[JAIR-23]*** [Mimicking Behaviors in Separated Domains](#)
Giuseppe De Giacomo, Dror Fried, Fabio Patrizi, **Shufang Zhu**
Journal of Artificial Intelligence Research 77 (2023):1087-1112

21. **[FMSD-23]*** [Finite-trace and generalized-reactivity specifications in temporal synthesis](#)
Giuseppe De Giacomo, Antonio Di Stasio, Lucas M. Tabajara, Moshe Y. Vardi, **Shufang Zhu**
Formal Methods System Design [\[Invited submission\]](#)
20. **[ECAI-23]*** [LTL_f Best-Effort Synthesis in Nondeterministic Planning Domains](#)
Giuseppe De Giacomo, Gianmarco Parretti, **Shufang Zhu**
In Proc. of the European Conference on Artificial Intelligence (ECAI) 2023
19. **[EUMAS-23]*** [LTL_f Synthesis Under Environment Specifications for Reachability and Safety Properties](#)
Benjamin Aminof, Giuseppe De Giacomo, Antonio Di Stasio, Hugo Francon, Sasha Rubin, **Shufang Zhu**
In Proc. of the European Conference on Multi-Agent Systems (EUMAS) 2023
18. **[EUMAS-23]*** [Symbolic LTL_f Best-Effort Synthesis](#)
Giuseppe De Giacomo, Gianmarco Parretti, **Shufang Zhu**
In Proc. of the European Conference on Multi-Agent Systems (EUMAS) 2023
17. **[VSTTE-22]*** [Compositional Safety LTL Synthesis](#)
Suguman Bansal, Giuseppe De Giacomo, Antonio Di Stasio, Yong Li, Moshe Y Vardi, **Shufang Zhu**
In Proc. of International Conference on Verified Software: Theories, Tools, and Experiments (VSTTE) 2022
16. **[IJCAI-22]*** [LTL_f Synthesis as AND-OR Graph Search: Knowledge Compilation at Work](#)
Giuseppe De Giacomo, Marco Favorito, Jianwen Li, Moshe Y Vardi, Shengping Xiao, **Shufang Zhu**
In Proc. of International Joint Conference on Artificial Intelligence (IJCAI) 2022
15. **[IJCAI-22]** [Synthesis of Maximally Permissive Strategies for LTL_f Specifications](#)
Shufang Zhu, Giuseppe De Giacomo
In Proc. of International Joint Conference on Artificial Intelligence (IJCAI) 2022
14. **[KR-22]** [Act for Your Duties but Maintain Your Rights](#)
Shufang Zhu, Giuseppe De Giacomo
In Proc. of International Conference on Principles of Knowledge Representation and Reasoning (KR) 2022
13. **[IJCAI-21]*** [Finite-Trace and Generalized-Reactivity Specifications in Temporal Synthesis](#)
Giuseppe De Giacomo, Antonio Di Stasio, Lucas M Tabajara, Moshe Y. Vardi, **Shufang Zhu**
In Proc. of International Joint Conference on Artificial Intelligence (IJCAI) 2021
12. **[AAAI-21]** [On-the-fly Synthesis for LTL over Finite Traces](#)
Shengping Xiao, Jianwen Li, **Shufang Zhu**, Yingying Shi, Geguang Pu, Moshe Y. Vardi
In Proc. of AAAI Conference on Artificial Intelligence (AAAI) 2021
11. **[KR-21]*** [Synthesis with Mandatory Stop Actions](#)
Giuseppe De Giacomo, Antonio Di Stasio, Giuseppe Perelli, **Shufang Zhu**
In Proc. of International Conference on Principles of Knowledge Representation and Reasoning (KR) 2021
10. **[GandALF-21]** [On the Power of Automata Minimization in Temporal Synthesis](#)
Shufang Zhu, Lucas M Tabajara, Geguang Pu, Moshe Y Vardi
In Proc. of International Symposium on Games, Automata, Logics, and Formal Verification (GandALF) 2021
9. **[KR-20]*** [Two-Stage Technique for LTL_f Synthesis Under LTL Assumptions](#)
Giuseppe De Giacomo, Antonio Di Stasio, Moshe Y. Vardi, **Shufang Zhu**
In Proc. of International Conference on Principles of Knowledge Representation and Reasoning (KR) 2020

8. [\[AAAI-20\] \$LTL_f\$ Synthesis with Fairness and Stability Assumptions](#)
Shufang Zhu, Giuseppe De Giacomo, Geguang Pu, Moshe Y Vardi
In Proc. of AAAI Conference on Artificial Intelligence (AAAI) 2020
7. [\[TAMC-19\] First-Order vs. Second-Order Encodings for \$LTL_f\$ -to-Automata Translation](#)
Shufang Zhu, Geguang Pu, Moshe Y. Vardi
In Proc. of Annual Conference of Theory and Applications of Models of Computation (TAMC) 2019
6. [\[FMSD-19\] SAT-based explicit LTL reasoning and its application to satisfiability checking](#)
Jianwen Li, **Shufang Zhu**, Geguang Pu, Lijun Zhang, Moshe Y. Vardi
Formal Methods System Design 54(2): 164-190
5. [\[FAC-18\] An explicit transition system construction approach to LTL satisfiability checking](#)
Jianwen Li, **Shufang Zhu**, Geguang Pu, Moshe Y. Vardi, Jifeng He
Formal Aspects of Computing 30(2): 193-217
4. [\[IJCAI-17\] Symbolic \$LTL_f\$ Synthesis](#)
Shufang Zhu, Lucas M. Tabajara, Jianwen Li, Geguang Pu, Moshe Y. Vardi
In Proc. of International Joint Conference on Artificial Intelligence (IJCAI) 2017
3. [\[HVC-17\] A Symbolic Approach to Safety LTL Synthesis](#)
Shufang Zhu, Lucas M. Tabajara, Jianwen Li, Geguang Pu, Moshe Y. Vardi
In Proc. of International Haifa Verification Conference (HVC) 2017
2. [\[ICCAD-17\] Safety model checking with complementary approximations](#)
Jianwen Li, **Shufang Zhu**, Yueling Zhang, Geguang Pu, Moshe Y. Vardi
In Proc. of IEEE/ACM International Conference on Computer-Aided Design (ICCAD) 2017
1. [\[HVC-15\] SAT-Based Explicit LTL Reasoning](#)
Jianwen Li, **Shufang Zhu**, Geguang Pu, Moshe Y. Vardi
In Proc. of International Haifa Verification Conference (HVC) 2015